

Audit of the repulsion coefficient in the CUDA phase-field cell simulation

Code review note

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Abstract

The interaction (repulsion) coefficient in the ϕ evolution equation is implemented at $30\kappa/\lambda^2$, whereas Palmieri *et al.* (2015) Eq. (S15) requires $60\kappa/\lambda^2$ — a factor of 2 too small. The velocity coefficient $60\kappa/(\xi\lambda^2)$ is correct. Every other term in the right-hand side matches the paper exactly. The missing factor is the double-counting of the ordered pair sum $\sum_{n,m \neq n}$ in the interaction free energy, not the mobility. The discrepancy has been confirmed by direct black-box measurement of the compiled CUDA binary (Sec. 6). Net effect: the simulation reproduces Palmieri’s model at $\kappa_{\text{eff}} = 5$, $\xi_{\text{eff}} = 750$ rather than the reported $\kappa = 10$, $\xi = 1500$.

1 Provenance and confidence

The model equations below were read from the local PDFs `papers/10.1038_srep11745.pdf` (main text) and `papers/srep11745_SI.pdf` (supplementary), extracted with `pypdf` in both default and `layout` mode. PDF mathematical typesetting does not extract cleanly, so the confidence in each reading is stated explicitly:

- **Eq. (S15) — high confidence.** The `layout`-mode extraction unambiguously shows a single prefactor $-30/\tilde{\lambda}^2$ applied to a bracket containing *both* the double-well term and the term $2\tilde{\kappa} \sum_{m \neq n} \phi_n \phi_m^2$. This is the operative equation and it is the basis of the whole finding.
- **Eq. (10) — high confidence.** Prefactor $30\kappa/\lambda^2$ and the ordered sum $\sum_{n,m \neq n}$ both extract cleanly.
- **Eq. (12) — high confidence.** Coefficient $60\kappa/(\xi\lambda^2)$ extracts cleanly.
- **Eq. (7) volume term — medium confidence.** The extraction is garbled, but shows two π and two R tokens, consistent with the normalised form $\frac{\mu_n}{\pi R^2}(\pi R^2 - \int \phi_n^2)^2$. This is independently confirmed by SI Table I, where $\mu = \gamma_0 \tilde{\mu}/(\pi \tilde{R}^2 \Delta^4)$, and by Eq. (S15), whose volume term is $-\frac{2\tilde{\mu}}{\pi \tilde{R}^2}$.

Crucially, the finding does *not* depend on the medium-confidence reading. Sec. 5 gives two independent cross-checks, and Sec. 6 gives a direct numerical measurement of the compiled binary.

2 Palmieri’s model as stated

All quantities are dimensionless; the paper states that Eq. (1) onward is written in dimensionless units.

2.1 Free energy

Single-cell part, Eq. (7):

$$\mathcal{F}_0 = \sum_n \left\{ \iint dx dy \left[\gamma_n (\nabla \phi_n)^2 + \frac{30\gamma_n}{\lambda^2} \phi_n^2 (1 - \phi_n)^2 \right] + \frac{\mu_n}{\pi R^2} \left(\pi R^2 - \iint dx dy \phi_n^2 \right)^2 \right\} \quad (1)$$

Total free energy, Eq. (9), and the interaction part, Eq. (10):

$$\mathcal{F} = \mathcal{F}_0 + \mathcal{F}_{\text{int}}, \quad \mathcal{F}_{\text{int}} = \iint dx dy \sum_{n, m \neq n} \frac{30\kappa}{\lambda^2} \phi_n^2 \phi_m^2. \quad (2)$$

Note the sum in Eq. (2) is *ordered*: it runs over all pairs (n, m) with $m \neq n$, so every unordered pair is counted twice. This is the origin of the factor under dispute.

2.2 Dynamics

Equation of motion, Eq. (1), with mobility $M = \frac{1}{2}$:

$$\frac{\partial \phi_n}{\partial t} + \mathbf{v}_n \cdot \nabla \phi_n = -\frac{1}{2} \frac{\delta \mathcal{F}}{\delta \phi_n}. \quad (3)$$

Velocity decomposition, Eq. (11), and the interaction velocity, Eq. (12):

$$\mathbf{v}_n = \mathbf{v}_{n,I} + \mathbf{v}_{n,A}, \quad \mathbf{v}_{n,I} = \frac{60\kappa}{\xi \lambda^2} \iint dx dy \phi_n (\nabla \phi_n) \sum_{m \neq n} \phi_m^2. \quad (4)$$

Run-and-tumble reorientation, Eq. (13):

$$P(t_r) dt_r = \frac{1}{\tau} e^{-t_r/\tau} dt_r. \quad (5)$$

2.3 The PDE that is actually integrated

SI Eq. (S15) is Eq. (3) with the functional derivative carried out. **This is the single most important equation in this document:**

$$\boxed{\frac{\partial \phi_n}{\partial t} + \mathbf{v}_n \cdot \nabla \phi_n = \gamma_n \nabla^2 \phi_n - \frac{30}{\lambda^2} \left[\gamma_n \phi_n (1 - \phi_n) (1 - 2\phi_n) + 2\kappa \sum_{m \neq n} \phi_n \phi_m^2 \right] - \frac{2\mu}{\pi R^2} \phi_n \left[\iint dx dy \phi_n^2 - \pi R^2 \right]} \quad (6)$$

Both the double-well term and the repulsion term sit inside *one* bracket sharing *one* prefactor $-30/\lambda^2$. Expanding:

$$\text{double-well contribution} = -\frac{30\gamma_n}{\lambda^2} \phi_n (1 - \phi_n) (1 - 2\phi_n), \quad (7)$$

$$\text{repulsion contribution} = -\frac{30}{\lambda^2} \cdot 2\kappa \cdot \phi_n \sum_{m \neq n} \phi_m^2 = -\frac{60\kappa}{\lambda^2} \phi_n \sum_{m \neq n} \phi_m^2. \quad (8)$$

2.4 Parameters (Table 1)

	γ_n	λ	R	κ	μ	τ	v_A	ξ
value	0.35 / 1	7	49	10	1	10^4	10^{-2}	1.5×10^3
cell type	cancer / normal					all		

3 Verification that Eq. (S15) follows from Eqs. (1)–(3)

This confirms the paper is internally self-consistent, and isolates where the factor of 2 comes from.

1. Gradient term. $\frac{\delta}{\delta\phi_n} \int \gamma_n (\nabla\phi_n)^2 = -2\gamma_n \nabla^2 \phi_n$. Times $-\frac{1}{2}$: $+\gamma_n \nabla^2 \phi_n$. **matches** Eq. (6).

2. Double well. Using $\frac{d}{d\phi} [\phi^2(1-\phi)^2] = 2\phi(1-\phi)(1-2\phi)$,

$$\frac{\delta}{\delta\phi_n} \int \frac{30\gamma_n}{\lambda^2} \phi_n^2 (1-\phi_n)^2 = \frac{60\gamma_n}{\lambda^2} \phi_n (1-\phi_n)(1-2\phi_n).$$

Times $-\frac{1}{2}$: $-\frac{30\gamma_n}{\lambda^2} \phi_n (1-\phi_n)(1-2\phi_n)$. **matches**

3. Volume constraint. With $V_n = \int \phi_n^2$ and $A_0 = \pi R^2$,

$$\frac{\delta}{\delta\phi_n} \frac{\mu}{A_0} (A_0 - V_n)^2 = \frac{\mu}{A_0} \cdot 2(A_0 - V_n) \cdot (-2\phi_n) = -\frac{4\mu}{A_0} (A_0 - V_n) \phi_n.$$

Times $-\frac{1}{2}$: $+\frac{2\mu}{A_0} (A_0 - V_n) \phi_n = -\frac{2\mu}{\pi R^2} \phi_n [V_n - \pi R^2]$. **matches**

4. Repulsion — where the factor of 2 lives. In the ordered sum of Eq. (2), index n appears in *both* slots. Collecting all ϕ_n -dependent terms:

$$\mathcal{F}_{\text{int}} \Big|_{\phi_n} = \frac{30\kappa}{\lambda^2} \left[\underbrace{\sum_{m \neq n} \phi_n^2 \phi_m^2}_{n \text{ in first slot}} + \underbrace{\sum_{m \neq n} \phi_m^2 \phi_n^2}_{n \text{ in second slot}} \right] = 2 \frac{30\kappa}{\lambda^2} \phi_n^2 \sum_{m \neq n} \phi_m^2.$$

Hence

$$\frac{\delta \mathcal{F}_{\text{int}}}{\delta \phi_n} = 2 \cdot \frac{30\kappa}{\lambda^2} \cdot 2\phi_n \sum_{m \neq n} \phi_m^2 = \frac{120\kappa}{\lambda^2} \phi_n \sum_{m \neq n} \phi_m^2,$$

and times $-\frac{1}{2}$:

$$\boxed{-\frac{60\kappa}{\lambda^2} \phi_n \sum_{m \neq n} \phi_m^2} \tag{9}$$

which is exactly the $2\tilde{\kappa}$ term printed inside the S15 bracket, Eq. (8). **matches**

5. Consistency of the velocity. Only $\mathcal{F}_{\text{int}}$ survives in $\int (\delta\mathcal{F}/\delta\phi_n) \nabla\phi_n$ (the single-cell terms are total derivatives and integrate to zero). With the same mobility $M = \frac{1}{2}$,

$$\mathbf{v}_{n,I} = \frac{1}{2\xi} \int \frac{\delta\mathcal{F}}{\delta\phi_n} \nabla\phi_n = \frac{1}{2\xi} \cdot \frac{120\kappa}{\lambda^2} \int \phi_n (\nabla\phi_n) \sum_{m \neq n} \phi_m^2 = \frac{60\kappa}{\xi\lambda^2} \int \phi_n (\nabla\phi_n) \sum_{m \neq n} \phi_m^2,$$

which is Eq. (4) exactly. **matches** The paper is internally consistent: Eq. (S15) and Eq. (12) both follow from the same $M \delta\mathcal{F}/\delta\phi$.

4 What the code implements

4.1 CUDA

Coefficient definitions, `include/types.cuh:26--36`:

```
template <typename T> __host__ __device__ inline T bulk_coeff(T lambda) {
    return T(30) / (lambda * lambda); }
template <typename T> __host__ __device__ inline T interaction_coeff(T kappa, T lambda) {
    return T(30) * kappa / (lambda * lambda); }
template <typename T> __host__ __device__ inline T motility_coeff(T kappa, T xi, T lambda) {
    return T(60) * kappa / (xi * lambda * lambda); }
```

Use, `src/kernels.cu:505--506, 546, 548`:

```
505: dwC_s = gam * bulk_coeff(lambda_); // 30*gamma/lambda^2
506: repC_s = interaction_coeff(kappa, lambda_); // 30*kappa /lambda^2
...
546: float rhs = gam*lap - dwC*dw + volC*c - repC*c*Soth;
547: float adv = vx*gx + vy*gy;
548: outp[idx] = c + dt * (rhs - adv);
```

There is no further scaling between line 506 and line 548: `rhs` is $\partial\phi/\partial t$ directly.

4.2 Reference implementations

Both CPU references use the same values, so neither is an independent oracle. From `tests/python/cpu_reference.py`:

```
132: two_keff = 60.0 * p.kappa / (p.lambd ** 2) # 12.245
165: repulsion = two_keff * phi * S
166: var_deriv = -tg * lap + bulk + constraint + repulsion
170: phi_new = phi + p.dt * (-0.5 * var_deriv - advection) # -> 6.122
```

`rust/cpu_ref/src/sim.rs`: lines 199, 342, 343, 345 are structurally identical. Note the trap: line 132 in isolation *looks* correct, because $60\kappa/\lambda^2$ is precisely the S15 coefficient — but it is placed in `var_deriv`, which line 170 multiplies by -0.5 .

4.3 Side-by-side

Term in $\partial\phi/\partial t$	Palmieri (S15)	Code	Status
Laplacian	$+\gamma$	$+\gamma$	matches
Double well	$-30\gamma/\lambda^2 = 0.612245$	0.612245	matches
Volume	$-2\mu/\pi R^2$	$-2\mu/\pi R^2$	matches
Repulsion	$-60\kappa/\lambda^2 = 12.244898$	6.122449	HALF
Velocity	$60\kappa/\xi\lambda^2 = 8.16327 \times 10^{-3}$	same	matches

(numerical values at $\gamma = 1$, $\kappa = 10$, $\lambda = 7$, $\mu = 1$, $\xi = 1500$.)

5 Two cross-checks independent of the free-energy convention

Both of the following hold regardless of how one normalises $\mathcal{F}_{\text{int}}$, so they cannot be evaded by a convention argument.

Check A: bulk/repulsion ratio. In Eq. (6) the double-well and repulsion terms share the single prefactor $-30/\lambda^2$. Their ratio is therefore fixed by the paper:

$$\frac{\text{repulsion coeff}}{\text{bulk coeff}} = \frac{2\kappa}{\gamma} = 20 \quad (\text{at } \kappa = 10, \gamma = 1), \quad \text{code: } \frac{30\kappa/\lambda^2}{30\gamma/\lambda^2} = \frac{\kappa}{\gamma} = 10.$$

In the code these two coefficients are two terms of the *same sum on the same line* (`kernels.cu:546`), multiplied by the same `dt` on line 548. Any global factor — the mobility, a units convention — scales both equally and cancels in the ratio. Since the bulk coefficient independently matches S15, the repulsion coefficient is short by 2.

Check B: repulsion/velocity ratio. The ϕ -equation repulsion coefficient and the velocity coefficient are the same object $M \delta\mathcal{F}_{\text{int}}/\delta\phi$, one of them divided by ξ . They must therefore differ by exactly ξ :

$$\text{Palmieri: } \frac{60\kappa/\lambda^2}{60\kappa/\xi\lambda^2} = \xi = 1500, \quad \text{code: } \frac{30\kappa/\lambda^2}{60\kappa/\xi\lambda^2} = \frac{\xi}{2} = 750.$$

Both checks independently demand $\text{repC} = 60\kappa/\lambda^2 = 12.2449$.

6 Direct measurement of the compiled binary

To remove any reliance on reading source code, the coefficient was measured from the binary’s own output. Sixteen cells at $\rho = 0.9$ were equilibrated to firm contact, then resumed with $\xi = 10^{12}$ so that the interaction velocity (hence all advection) is $\sim 10^{-11}$ and the dynamics is pure gradient flow. Two checkpoints one `dt` apart give $\partial\phi/\partial t$ by finite difference. Every term except the repulsion is independently verified, so one solves pointwise:

$$C = - \frac{\left. \frac{\partial\phi}{\partial t} \right|_{\text{measured}} - \left[\gamma \nabla^2 \phi - \frac{30\gamma}{\lambda^2} \phi(1-\phi)(1-2\phi) + \frac{2\mu}{A_0} (A_0 - V) \phi \right]}{\phi S_{\text{other}}}. \quad (10)$$

```
cell 0: 1494 px C = 6.1224 (IQR 6.1224..6.1225)
cell 1: 1738 px C = 6.1224 (IQR 6.1224..6.1225)
cell 2: 1379 px C = 6.1224 (IQR 6.1224..6.1225)
cell 3: 1176 px C = 6.1224 (IQR 6.1224..6.1225)

pooled over 16 cells, 23265 pixels
MEASURED repulsion coefficient C = 6.12245
  candidate 30k/l^2 = 6.12245 ratio C/cand = 1.00000
  candidate 60k/l^2 = 12.24490 ratio C/cand = 0.50000
```

Agreement with $30\kappa/\lambda^2$ to six significant figures over 2.3×10^4 pixels. **The binary evaluates half of the Palmieri coefficient.**

7 Consequences and remedy

7.1 Equivalent parameter set

Since κ and ξ enter *only* through `interaction_coeff` and `motility_coeff` (verified by exhaustive grep), the implemented model is exactly Palmieri’s model evaluated at

$$\kappa_{\text{eff}} = 5, \quad \xi_{\text{eff}} = 750, \quad (11)$$

while the runs are labelled $\kappa = 10$, $\xi = 1500$. Check: $60 \cdot 5/\lambda^2 = 6.1224 \checkmark$ and $60 \cdot 5/(750\lambda^2) = 8.163 \times 10^{-3} \checkmark$.

7.2 Zero-code-change workaround

By the same token, running the *current unmodified binary* with

```
--kappa 20 --xi 3000
```

gives repulsion $30 \cdot 20/\lambda^2 = 12.2449 \checkmark$ and motility $60 \cdot 20/(3000\lambda^2) = 8.163 \times 10^{-3} \checkmark$ — i.e. exactly Palmieri’s intended physics. This permits a paired A/B comparison quantifying the impact on observables before any code is touched, and serves as a validation target for any patch.

7.3 Code fix

File	Current	Fixed
src/kernels.cu:506	<code>interaction_coeff(kappa, lambda_)</code>	<code>2.0f * interaction_coeff(...)</code>
cpu_reference.py:132	<code>60.0 * kappa / lambda**2</code>	<code>120.0 * kappa / lambda**2</code>
rust/cpu_ref/sim.rs:199	<code>60.0 * kappa / lambda**2</code>	<code>120.0 * kappa / lambda**2</code>

`motility_coeff` must **not** be changed.

7.4 Why no existing test caught this

The two tests that act as physics oracles — `test_isolated_cell_energy_decreases` and `test_single_cell_pde_residual_small` — are both **single-cell**. The latter states in a comment “No interaction term (single cell)”. The repulsion coefficient is the only coefficient that requires ≥ 2 cells to exercise, so no oracle test ever evaluates it. All other physics tests compare the CUDA code against `cpu_reference.py` or the Rust reference, which encode the same derived constants — circular by construction.

Recommended additions: (i) a convention-free assertion that $\text{repC}/\text{motility_coeff} = \xi$; and (ii) a two-cell variational test that writes $\mathcal{F}$ explicitly and differentiates it *numerically*, asserting $\partial_t \phi = -\frac{1}{2} \delta \mathcal{F} / \delta \phi$ in the overlap region.

7.5 Impact on motility results

- **Dilute / isolated cells: unaffected.** $\mathbf{v}_{n,I} \rightarrow 0$, so $\mathbf{v} = v_A \hat{\mathbf{p}}$ and Palmieri Eq. (14), $\langle \Delta x^2 \rangle = 2v_A^2 \tau^2 [t/\tau - 1 + e^{-t/\tau}]$ with $D_{\text{eff}} = v_A^2 \tau / 2$, holds exactly.
- **Confluent monolayer: affected.** Velocities are correct for a given configuration, but the configurations visited are not: cells interpenetrate roughly twice as far before generating the same restoring force. Palmieri chose κ large specifically so cells “deform when brought together rather than overlap”; halving it moves the system toward the regime he tuned against.

- Deformation observables (L_n , and hence $p_{\text{eff}} = 2\sqrt{\pi}L$) are compressed toward unity; the soft/normal contrast shrinks in magnitude.
- Caging is weaker, so D_{eff} in the monolayer runs high and the jamming/glass transition ρ_c shifts upward — the principal risk for any finite-size-scaling analysis.
- Speed bursts, whose mechanism is neighbour pinching during T1 events with force $\propto \kappa$, are weakened.
- Existing equilibration checkpoints relaxed into a more-overlapped, less-deformed packing, so they are not valid initial conditions for a corrected binary.
- Qualitative comparative conclusions driven by the elasticity contrast $\gamma_c/\gamma_n = 0.35$ — which is implemented correctly — are likely to survive; quantitative comparison against Palmieri’s published numbers is not.

8 Manuscript equations

For completeness, the free energy as typeset in `nibi_sync/manuscript.tex` does not reproduce Eqs. (1)–(2) either, in two of three terms:

	Manuscript (as typeset)	vs. Palmieri
F_{int} (line 278)	$\gamma_i \nabla \phi_i ^2 + \frac{30\gamma_i}{\lambda^2} \phi_i^2 (1 - \phi_i)^2$	identical matches
F_{vol} (line 289)	$\mu \sum_i \left(\int \phi_i^2 - A_0 \right)^2$	missing $1/A_0$
F_{rep} (line 298)	$\kappa \sum_{i < j} \int \phi_i^2 \phi_j^2$	missing $60/\lambda^2$

Since $\sum_{n,m \neq n} = 2 \sum_{i < j}$, Palmieri’s Eq. (10) equals $\frac{60\kappa}{\lambda^2} \sum_{i < j}$; matching it with the manuscript’s form would require $\kappa = 12.24$, whereas the manuscript’s parameter table (line 369) quotes Palmieri’s $\kappa = 10$. The manuscript states the model is used “without modification”, so these two equations should be corrected to

$$F_{\text{rep}} = \frac{30\kappa}{\lambda^2} \sum_{i, j \neq i} \int \phi_i^2 \phi_j^2, \quad F_{\text{vol}} = \frac{\mu}{A_0} \sum_i \left(\int \phi_i^2 - A_0 \right)^2.$$

Note that the manuscript’s error and the code’s error are in *different* directions ($-10\phi S$ implied by the manuscript, -6.12 evaluated by the code, -12.24 required by Palmieri), so they are independent fixes.